

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022**(CBCSS-UG)****Mathematics****MTS 6B 10—REAL ANALYSIS****(2019 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 6B 10—REAL ANALYSIS

(Multiple Choice Questions for SDE Candidates)

1. Which of the following are true about Thomae's function ?
- (A) Continuous only at irrationals. (B) Continuous every where on $\mathbb{R}$.
(C) Continuous only at rationals. (D) Discontinuous only at irrationals.
2. $g : A \rightarrow \mathbb{R}$, where $A \subseteq \mathbb{R}$, is uniformly continuous function when, :
- (A) $A = (0, 1)$. (B) $A = (-1, 0)$.
(C) $A = [a, 1) ; a > 0$. (D) $A = (0, \infty)$.
3. Let f and g be *two* Lipschitz functions on $A \subset \mathbb{R}$. Choose the false statement from below :
- (A) cf is a Lipschitz function on A where c is a constant.
(B) $f + g$ is Lipschitz function on A .
(C) f^2 is a Lipschitz function on A .
(D) $f - g$ is a Lipschitz function on A .
4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = \frac{1}{1+x^2}$ then,
- (A) f is uniformly continuous on $\mathbb{R}$.
(B) f is not uniformly continuous on $\mathbb{R}$.
(C) f is uniformly continuous only on bounded subset of $\mathbb{R}$.
(D) f is uniformly continuous only on a closed bounded subset of $\mathbb{R}$.

5. If $0 < x < 1$, which of the following terms increases as x increases :

(i) $1 - x^3$.

(ii) $x - 1$.

(iii) $\frac{1}{x^2}$.

(A) (i) only.

(B) (ii) only.

(C) (iii) only.

(D) (i) and (iii).

6. Evaluate $\lim_{x \rightarrow 0} \frac{\log x}{\cot x}$.

(A) 0.

(B) 1.

(C) 2.

(D) 0.5.

7. Let $f(x) = g(x)$ for $[0, 1] \setminus \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$. Choose the false statement from below :

(A) If f is Riemann integrable on $[0, 1]$ then g also Riemann integrable on $[0, 1]$.

(B) If g is Riemann integrable on $[0, 1]$ the f also Riemann integrable on $[0, 1]$.

(C) $\int f dx = \int g dx$.

(D) If f is Riemann integrable on $[0, 1]$ then $\int f dx = \int g dx$.

8. Which of the following is true ?

(A) There exists $f \in \mathcal{R}[a, b]$, such that f is not bounded on $[a, b]$.

(B) There exists $f \in \mathcal{R}[a, b]$, such that f is unbounded on $[a, b]$.

(C) Unbounded functions on $[a, b]$ is integrable when functions takes only positive values.

(D) If f is not bounded on $[a, b]$ then $f \notin \mathcal{R}[a, b]$.

Turn over

9. Let $H(x) := k$ for $x = 1/k$ ($k \in \mathbb{N}$) $H(x) := 0$ elsewhere on $[0, 1]$. H is not Riemann integrable.

(A) $H \notin \mathcal{R}[0, 1]$.

(B) $\int_0^1 H = 0$.

(C) $\int_0^1 H = 1$.

(D) $\int_0^1 H = \frac{1}{2}$.

10. Choose the false statement from below :

(A) Any subset of a null set is also a null set.

(B) Union of two null set is a null set.

(C) Countable set is a null set.

(D) Every null set is a countable set.

11. Let $g_n(x) := x^n$ for $x \in \mathbb{R}$, $n \in \mathbb{N}$ then :

(A) (g_n) converges to a continuous function on $(-1, 1]$.

(B) (g_n) converges to a function on $(-2, 1]$.

(C) (g_n) converges to a continuous function on $(0, 1]$.

(D) (g_n) converges to a continuous function on $(-1, 1)$.

12. For $x \in \mathbb{R}$, $\lim ((1/n) \sin(nx + n)) = ?$

(A) -1 .

(B) 0 .

(C) 1 .

(D) 2 .

13. Let (f_n) converges uniformly on $\mathbb{R}$ to f , then :

(A) $(2f_n)$ does not converge uniformly on $\mathbb{R}$.

(B) (f_n^2) converge uniformly on $\mathbb{R}$.

(C) (f_n^3) converge uniformly on $\mathbb{R}$.

(D) None of these.

14. Find $\sum_{n=0}^{\infty} \frac{1}{(\alpha+n)(\alpha+n+1)}$ if $\alpha > 0$.

(A) $\frac{1}{\alpha}$

(B) 0.

(C) $\frac{1}{\alpha^2}$

(D) 2α .

15. Choose the correct statement :

(A) $\sum_{n=1}^{\infty} \cos n$ is divergent and the series $\sum_{n=1}^{\infty} (\cos n)/n^2$ is convergent.

(B) $\sum_{n=1}^{\infty} \cos n$ is divergent and the series $\sum_{n=1}^{\infty} (\cos n)/n^2$ is divergent.

(C) $\sum_{n=1}^{\infty} \cos n$ is convergent and the series $\sum_{n=1}^{\infty} (\cos n)/n^2$ is convergent.

(D) $\sum_{n=1}^{\infty} \cos n$ is convergent and the series $\sum_{n=1}^{\infty} (\cos n)/n^2$ is divergent.

16. Find $\int_{-\infty}^0 e^x dx$:

(A) 1.

(B) 0.

(C) -1.

(D) -2.

17. Find $\int_{-\infty}^0 x^2 dx$.

(A) 1.

(B) ∞ .

(C) 0.

(D) None of these.

18. Find P.V. $\int_{-\infty}^{\infty} x dx$:

(A) 1.

(B) ∞ .

(C) 0.

(D) $-\infty$.

19. Find P.V. $\int_{-\infty}^{\infty} x^2 dx$:

(A) 1.

(B) ∞ .

(C) 0.

(D) $-\infty$.

20. Find P.V. $\int_{-\infty}^{\infty} \frac{dx}{x^2 + 1}$:

(A) π .

(B) $\frac{\pi}{2}$.

(C) 0.

(D) $-\pi$.

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SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022

(CBCSS-UG)

Mathematics

MTS 6B 10—REAL ANALYSIS

(2019 Admissions)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A*Answer at least ten questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 30.*

1. Define continuity of a function. Show that the constant function $f(x) = b$ is continuous on $\mathbb{R}$.
2. State Boundedness theorem. Is boundedness of the interval, a necessary condition in the theorem? Justify with an example.
3. If $f : A \rightarrow \mathbb{R}$ is uniformly continuous on $A \subseteq \mathbb{R}$ and (x_n) is a Cauchy sequence in A . Then show that $f(x_n)$ is a Cauchy sequence in $\mathbb{R}$.
4. Define Riemann sum of a function $f : [a, b] \rightarrow \mathbb{R}$.
5. Suppose f and g are in $\mathbb{R}[a, b]$ then show that $f + g$ is in $\mathbb{R}[a, b]$.
6. State squeeze theorem for Riemann integrable functions.
7. If f belong to $\mathbb{R}[a, b]$, then show that its absolute value $|f|$ is in $\mathbb{R}[a, b]$.
8. Define pointwise convergence of a sequence (f_n) of functions.
9. Discuss the uniform convergence of $f_n(x) = x^n$ on $(-1, 1]$.
10. If $h_n(x) = 2nxe^{-nx^2}$ for $x \in [0, 1], n \in \mathbb{N}$ and $h(x) = 0$ for all $x \in [0, 1]$, then show that :

$$\lim_{n \rightarrow \infty} \int_0^1 h_n(x) dx \neq \int_0^1 h(x) dx.$$
11. State Cauchy criteria for uniform convergence series of functions.